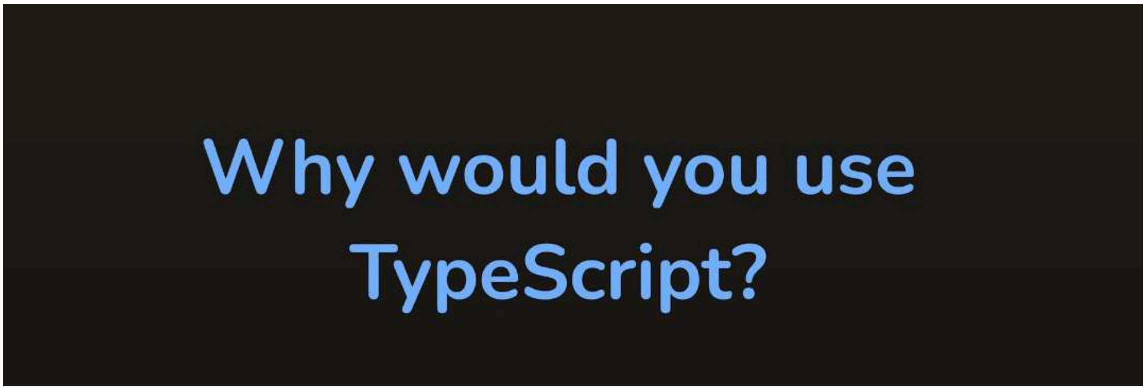
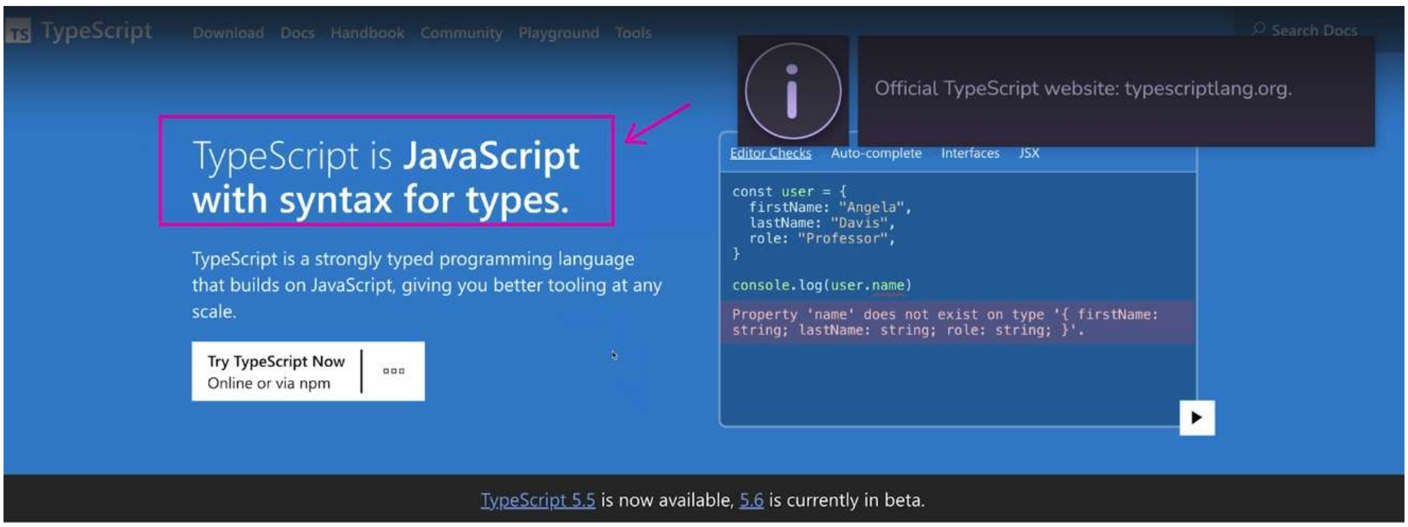
# Section 1 -Getting Started

2026年6月1日 9:46




## What is TypeScript?

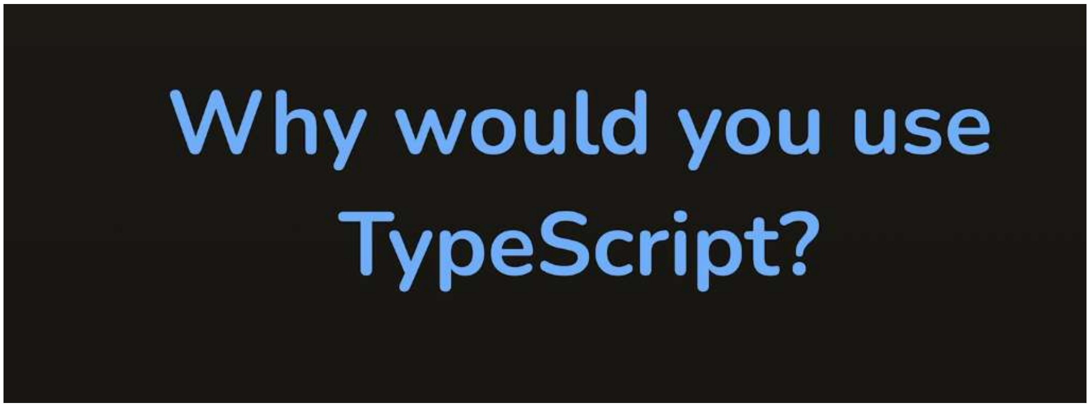
JavaScript and More

A Result You Can Trust

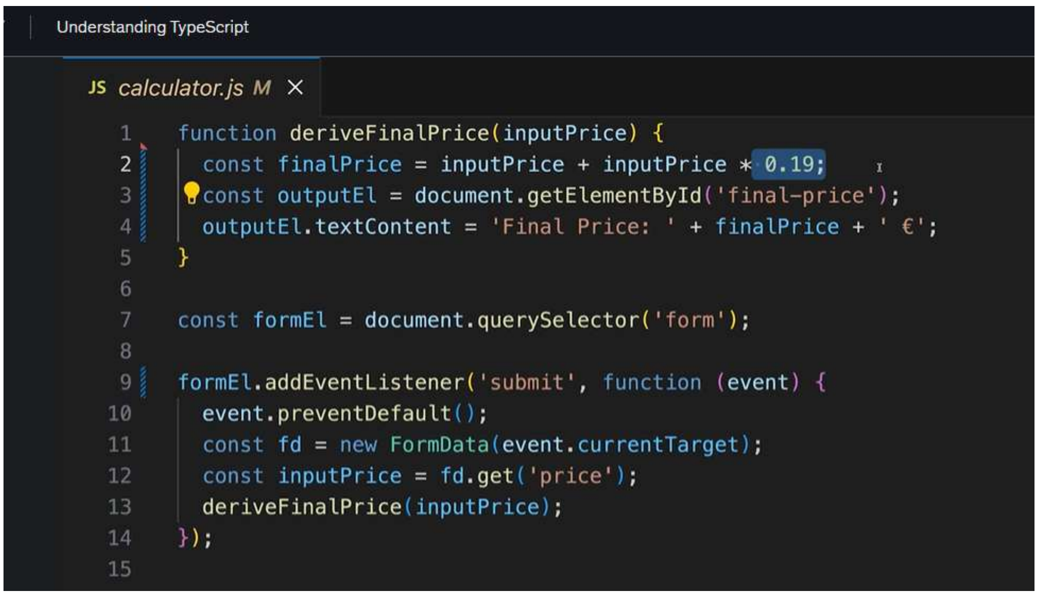
Safety at Scale

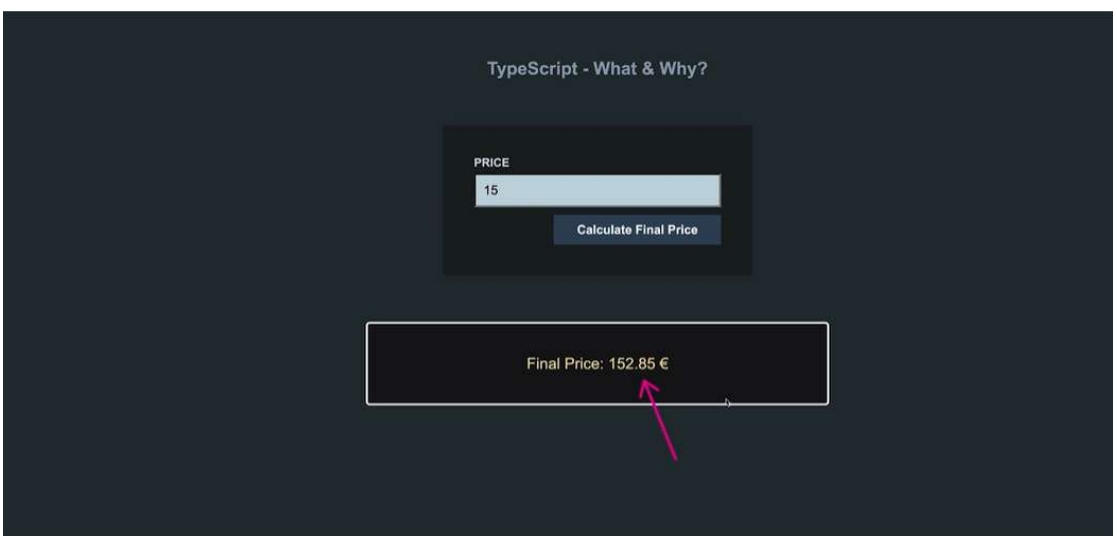
```typescript
function deriveFinalPrice(inputPrice: number) {
  const finalPrice = inputPrice + inputPrice * 0.19;
  const outputEl = document.getElementById('final-price')!;
  outputEl.textContent = 'Final Price: ' + finalPrice + ' €';
}

const formEl = document.querySelector('form')!;

formEl.addEventListener('submit', function (event) {
  event.preventDefault();
  const fd = new FormData(event.currentTarget as HTMLFormElement);
  const inputPrice = fd.get('price')!;
  deriveFinalPrice(+inputPrice);
});
```


## 2. Why would you use TypeScript?




I have prepared a project



JS calculator.js M

```javascript
function deriveFinalPrice(inputPrice) {
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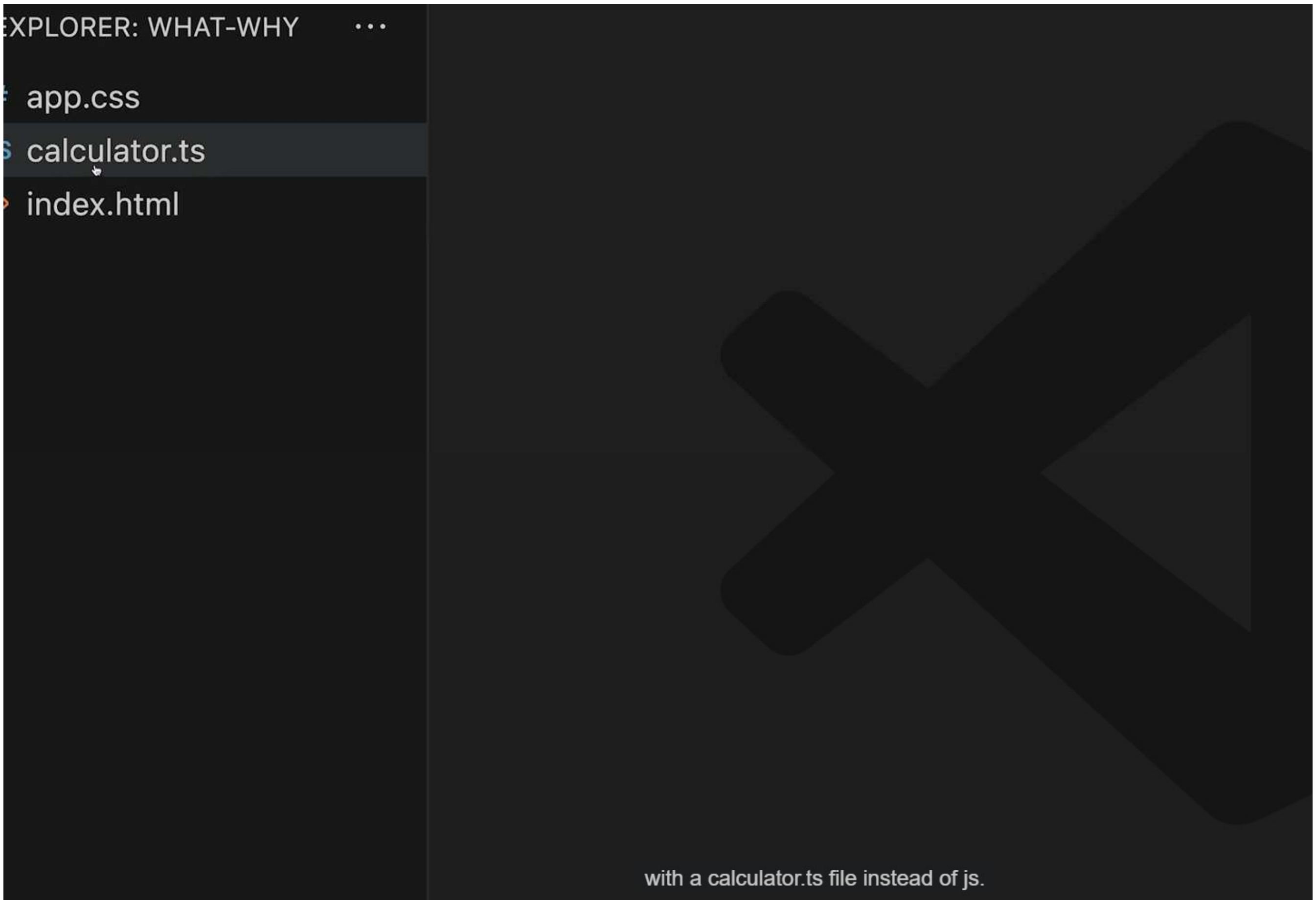
I'm having some problems with the types

Problems could have been avoided if we were using typescript.




calculator.ts

```ts
function deriveFinalPrice(inputPrice: number) {
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TS calculator.ts 1

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});
```

```
> 2.85 + "15"
<· '2.8515'
```

and produce a longer string where it simply concatenates

## Installing and Using TypeScript

This typescript code does not run on the browser.

calculator.ts

```typescript
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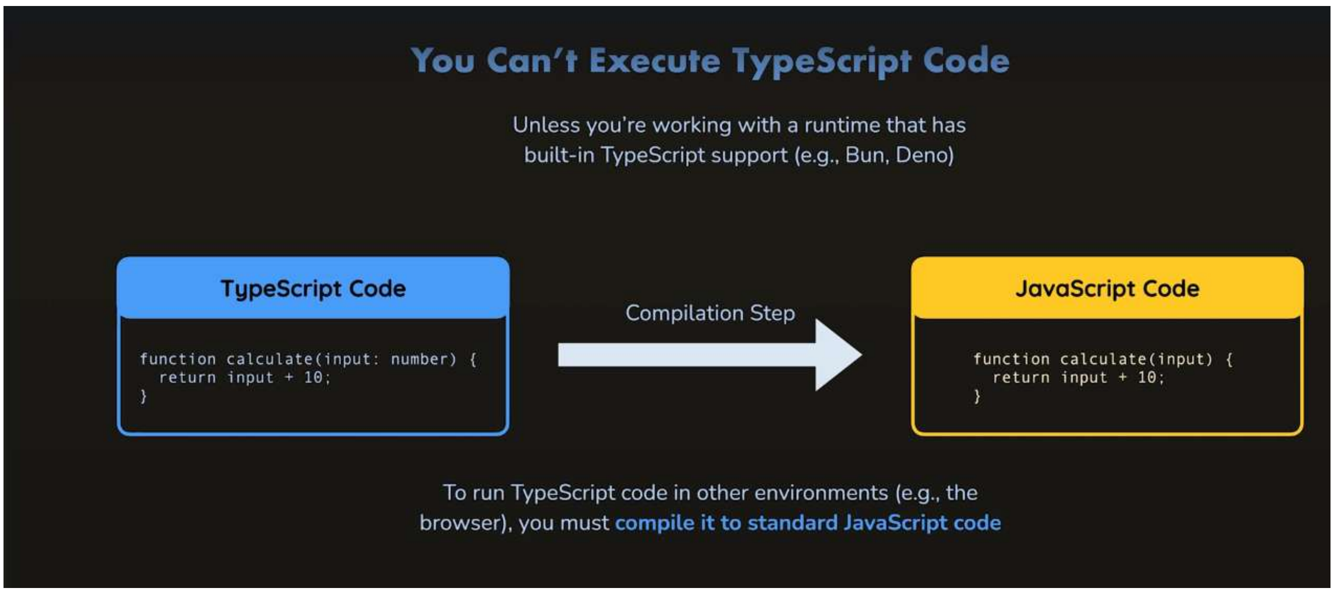
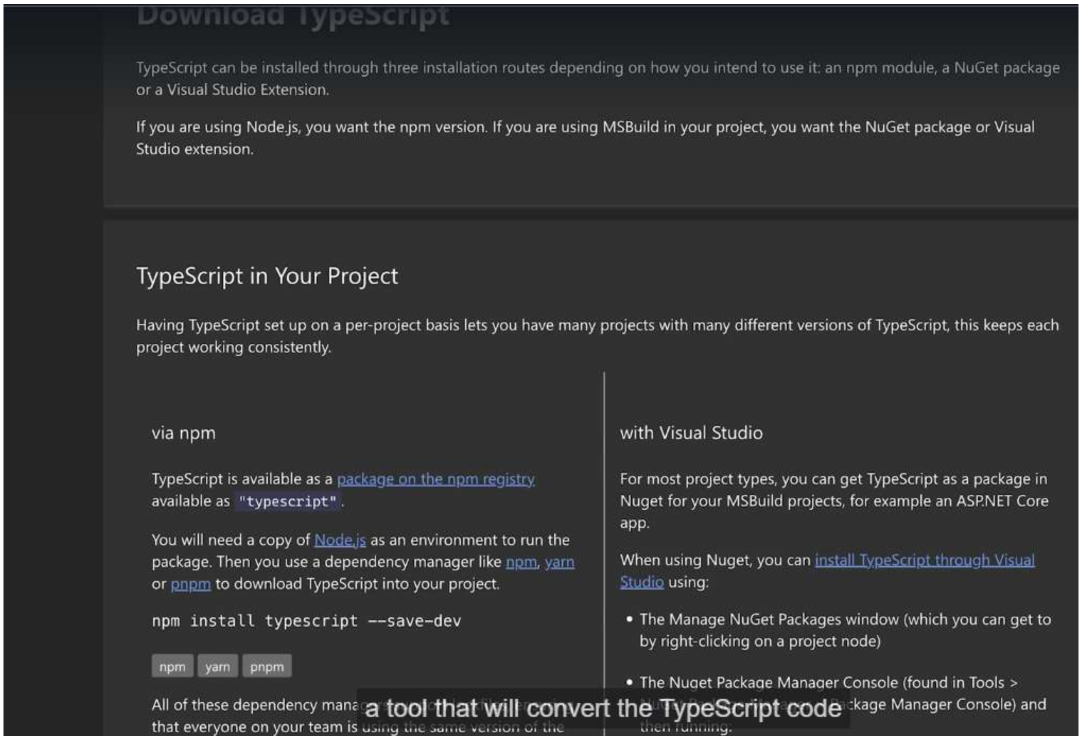



That's why in the official doc of typescript, you need to download a compiler that will compile your typescript code to javascript code.


## Download TypeScript

TypeScript can be installed through three installation routes depending on how you intend to use it: an npm module, a NuGet package or a Visual Studio Extension.

If you are using Node.js, you want the npm version. If you are using MSBuild in your project, you want the NuGet package or Visual Studio extension.

### TypeScript in Your Project

Having TypeScript set up on a per-project basis lets you have many projects with many different versions of TypeScript, this keeps each project working consistently.

#### via npm

TypeScript is available as a package on the npm registry available as `"typescript"`.

You will need a copy of Node.js as an environment to run the package. Then you use a dependency manager like npm, yarn or pnpm to download TypeScript into your project.

```
npm install typescript --save-dev
```

npm yarn pnpm

All of these dependency mana… that everyone on your team is using the same version of the

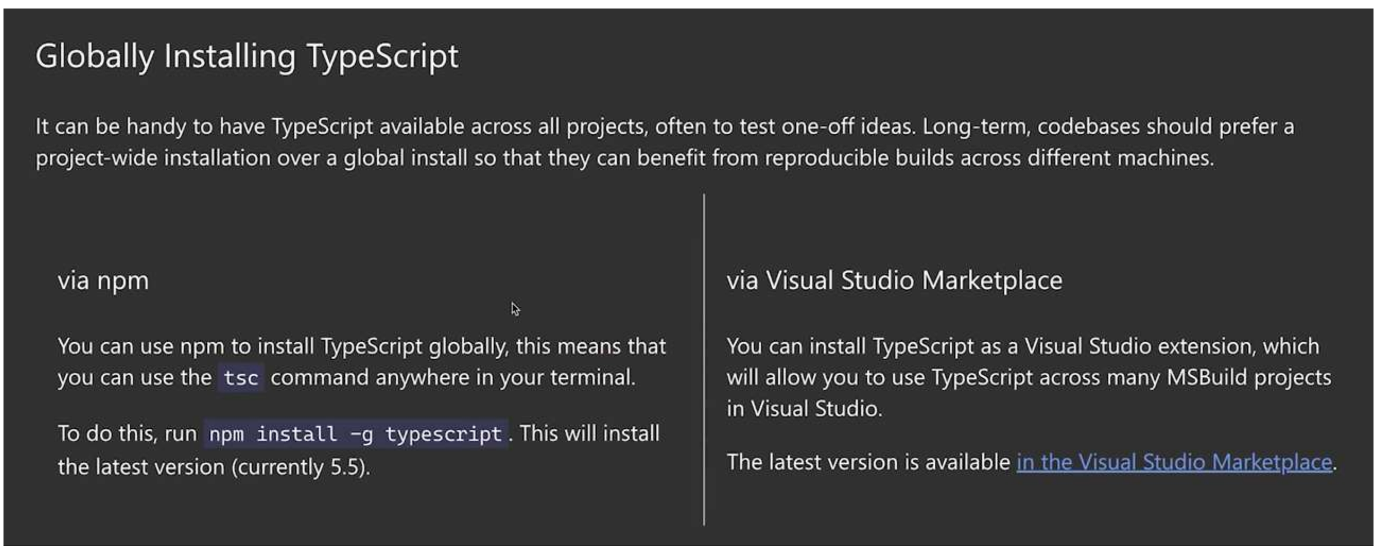
#### with Visual Studio

For most project types, you can get TypeScript as a package in Nuget for your MSBuild projects, for example an ASP.NET Core app.

When using Nuget, you can install TypeScript through Visual Studio using:

- The Manage NuGet Packages window (which you can get to by right-clicking on a project node)
- The Nuget Package Manager Console (found in Tools > …ckage Manager Console) and then running:




### Globally Installing TypeScript

It can be handy to have TypeScript available across all projects, often to test one-off ideas. Long-term, codebases should prefer a project-wide installation over a global install so that they can benefit from reproducible builds across different machines.

#### via npm

You can use npm to install TypeScript globally, this means that you can use the `tsc` command anywhere in your terminal.

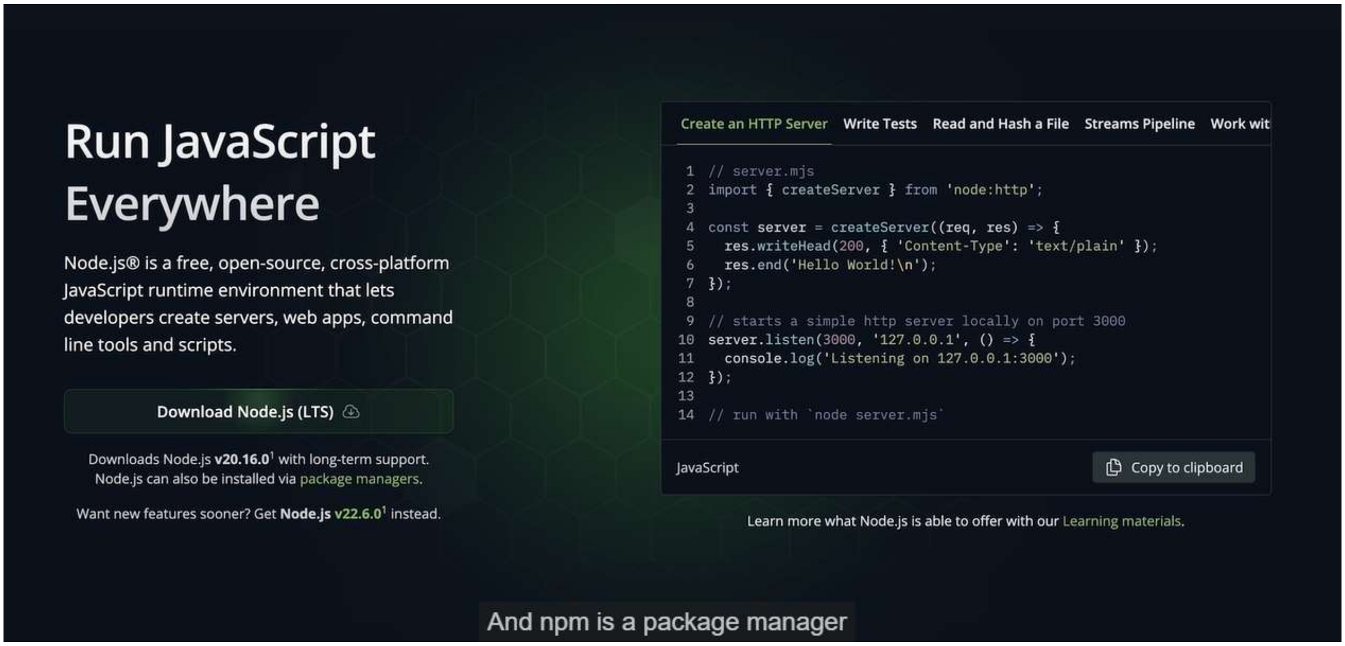
To do this, run `npm install -g typescript`. This will install the latest version (currently 5.5).

#### via Visual Studio Marketplace

You can install TypeScript as a Visual Studio extension, which will allow you to use TypeScript across many MSBuild projects in Visual Studio.

The latest version is available in the Visual Studio Marketplace.

We will install via npm.

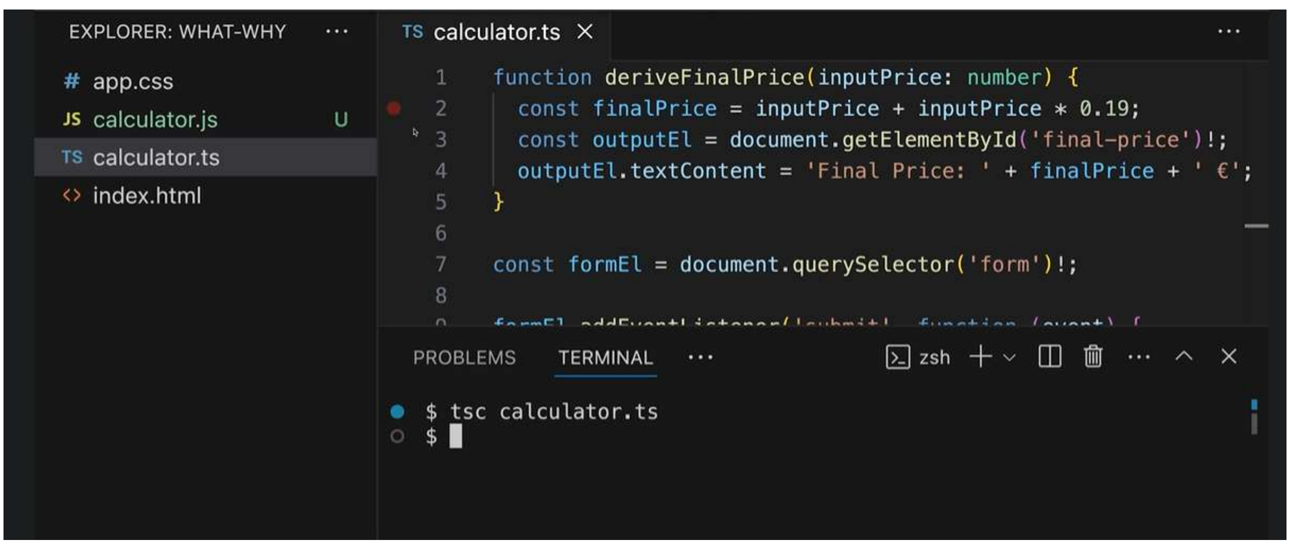


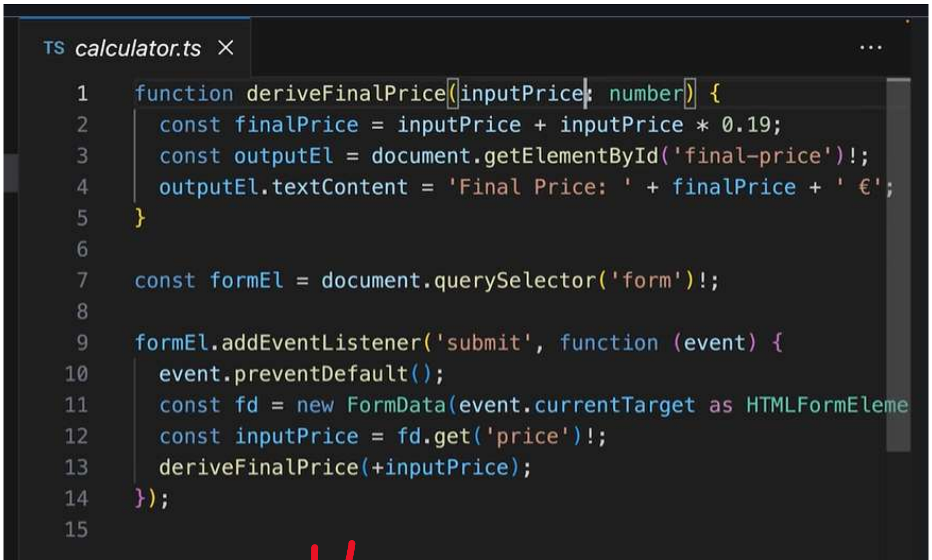

Npm is a package manager that comes together with Node.JS.

Now open up the terminal and type

'tsc calculator.ts'

And you will see the .js file





TS calculator.ts

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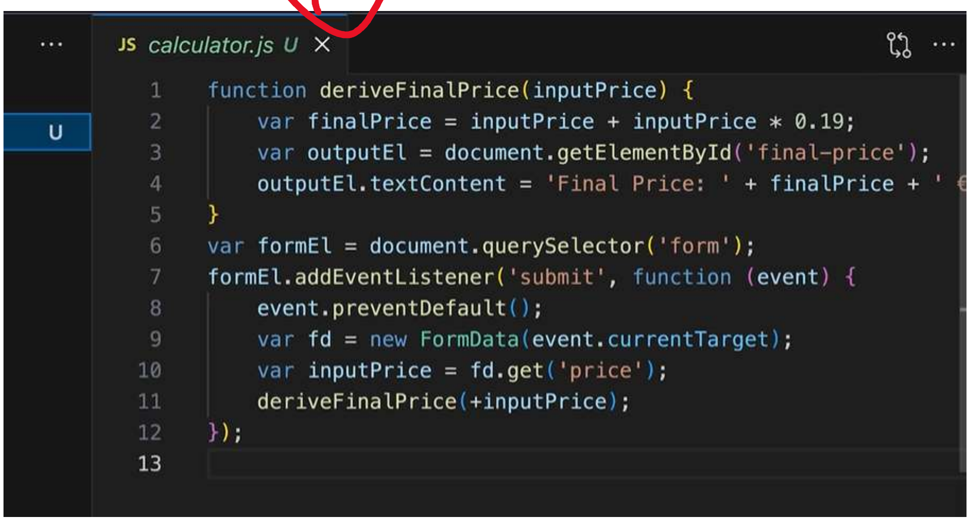

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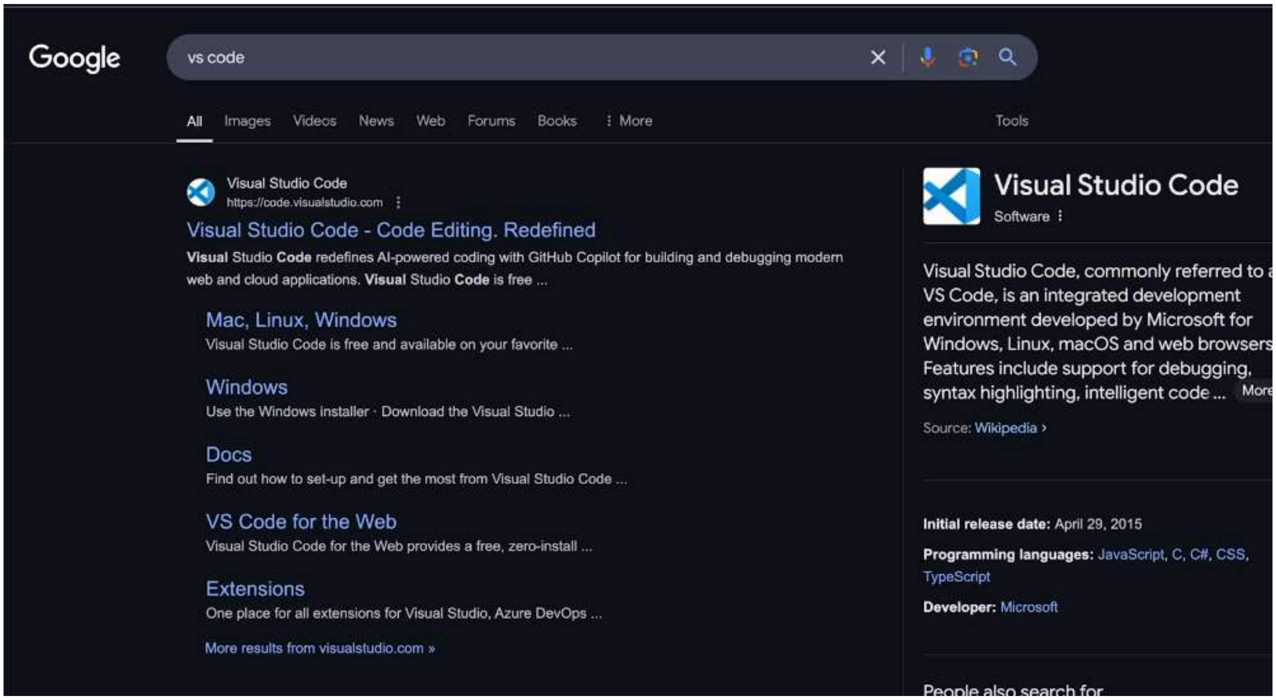
# Editor and Project Setup




## About this Course